# Olist KPI Analysis SQL

## STEP 4 — SETUP IN MYSQL WORKBENCH

### STEP 4.1 — Create Database

```
CREATE DATABASE ecommerce_db;
USE ecommerce_db;
```

### STEP 4.2 — Create Table

```
CREATE TABLE master_data (
order_id VARCHAR(50),
customer_unique_id VARCHAR(50),
order_date DATE,
product_id VARCHAR(50),
product_category_name TEXT,
price FLOAT,
freight_value FLOAT,
payment_value FLOAT,
customer_city TEXT,
customer_state VARCHAR(10),
revenue FLOAT,
order_month VARCHAR(10),
order_year INT
);
```

### STEP 4.3 — Import CSV

In MySQL Workbench:

1. Right-click on the master_data table
2. Click → Table Data Import Wizard
3. Select:

master_dataset.csv

4. Map columns (auto usually works)
5. Finish

```
SELECT COUNT(*) FROM master_data;
SELECT * FROM master_data LIMIT 10;
```

>> to check

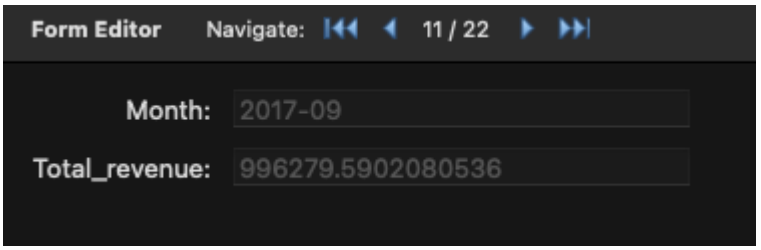
STEP 4.4 — Now Write Queries

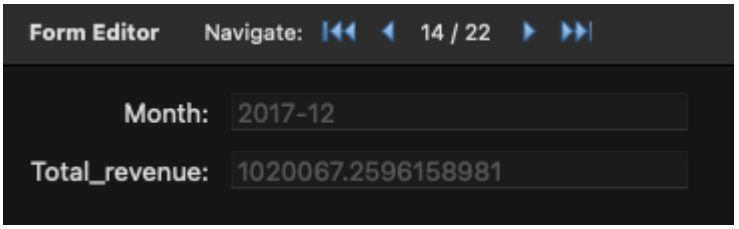
### Total Revenue

```
SELECT SUM(revenue) AS total_revenue
FROM master_data;
```

### Monthly Revenue

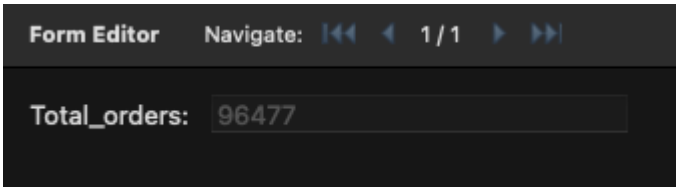
```
SELECT
  DATE_FORMAT(order_date, '%Y-%m') AS month,
  SUM(revenue) AS total_revenue
FROM master_data
GROUP BY month
ORDER BY month;
```




### Total Orders

```
SELECT COUNT(DISTINCT order_id) AS total_orders
FROM master_data;
```




### Total Customers

```
SELECT COUNT(DISTINCT customer_unique_id) AS total_customers
FROM master_data;
```

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Total_customers: 96477

## Average Order Value

```
SELECT
  SUM(revenue) / COUNT(DISTINCT order_id) AS avg_order_value
FROM master_data;
```

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Avg_order_value: 204.98316115294688

## MOM GROWTH (WINDOW FUNCTION in MySQL)

```
SELECT
  month,
  total_revenue,
  LAG(total_revenue) OVER (ORDER BY month) AS prev_month,
  (total_revenue - LAG(total_revenue) OVER (ORDER BY month))
     / LAG(total_revenue) OVER (ORDER BY month) AS growth_rate
FROM (
  SELECT
     DATE_FORMAT(order_date, '%Y-%m-01') AS month,
     SUM(revenue) AS total_revenue
  FROM master_data
  GROUP BY month
) t;
```

Result Grid Filter Rows: Search Export:

| month | total_revenue | prev_month | growth_rate |
|---|---|---|---|
| 2016-10-01 | 61746.94012451172 | NULL | NULL |
| 2016-12-01 | 19.6200008392334 | 61746.94012451172 | -0.9996822514476075 |
| 2017-01-01 | 176491.48991680145 | 19.6200008392334 | 8994.488397935125 |
| 2017-02-01 | 325782.6590719223 | 176491.48991680145 | 0.8458831030634797 |
| 2017-03-01 | 505735.8299264908 | 325782.6590719223 | 0.5523718523484721 |
| 2017-04-01 | 456108.3195180893 | 505735.8299264908 | -0.09812931469699289 |
| 2017-05-01 | 701313.6000938416 | 456108.3195180893 | 0.5376031746906721 |
| 2017-06-01 | 585400.9801893234 | 701313.6000938416 | -0.16527929857485732 |
| 2017-07-01 | 716069.9797058105 | 585400.9801893234 | 0.22321281299226337 |
| 2017-08-01 | 842689.9406518936 | 716069.9797058105 | 0.176826238404944 |
| 2017-09-01 | 996279.5902080536 | 842689.9406518936 | 0.18226116409713528 |
| 2017-10-01 | 998609.6187181473 | 996279.5902080536 | 0.002338729542383888 |
| 2017-11-01 | 1548682.6884908676 | 998609.6187181473 | 0.5508389459324603 |
| 2017-12-01 | 1020067.2596158981 | 1548682.6884908676 | -0.3413323031266561 |
| 2018-01-01 | 1374064.0189962387 | 1020067.2596158981 | 0.3470327628333415 |
| 2018-02-01 | 1279970.4486961365 | 1374064.0189962387 | -0.06847830159240914 |
| 2018-03-01 | 1435458.330400467 | 1279970.4486961365 | 0.12147771213212055 |
| 2018-04-01 | 1466607.1519851685 | 1435458.330400467 | 0.021699565166765643 |
| 2018-05-01 | 1480667.5905733109 | 1466607.1519851685 | 0.00958705169895734 |
| 2018-06-01 | 1285926.1116991043 | 1480667.5905733109 | -0.13152275373218855 |
| 2018-07-01 | 1307228.1779966354 | 1285926.1116991043 | 0.01656554455479914 |
| 2018-08-01 | 1211240.0919733047 | 1307228.1779966354 | -0.07342871553643769 |

## TOP CUSTOMERS

```
SELECT
  customer_unique_id,
  SUM(revenue) AS total_spend
FROM master_data
GROUP BY customer_unique_id
ORDER BY total_spend DESC
LIMIT 10;
```

Result Grid  Filter Rows: Search

| customer_unique_id | total_spend |
|---|---|
| 1617b1357756262bfa56ab541c47bc16 | 109312.640625 |
| bd5d39761aa56689a265d95d8d32b8be | 45256.0009765625 |
| be1b70680b9f9694d8c70f41fa3dc92b | 44047.998046875 |
| 05455dfa7cd02f13d132aa7a6a9729c6 | 36489.240234375 |
| 1ff773612ab8934db89fd5afa8afe506 | 30186.0009765625 |
| ec5b2ba62e574342386871631fafd3fc | 29099.51953125 |
| e7d6802668de6e74d0d6c56565bf2a24 | 22346.59912109375 |
| 8c20d9bfbc96c5d39025d77a3ba83d7f | 21874.050537109375 |
| f7622098214b4634b7fe7eee269b5426 | 19457.04052734375 |
| 71901689c5f3e5adc27b1dd16b33f0b8 | 19174.3798828125 |

## REPEAT CUSTOMERS

```
SELECT
  customer_unique_id,
  COUNT(DISTINCT order_id) AS order_count
FROM master_data
GROUP BY customer_unique_id
HAVING COUNT(DISTINCT order_id) > 1;
```

Result Grid  Filter Rows:

| customer_unique... | order_count |
|---|---|

## CUSTOMER RANKING (Window Function)

```
SELECT
  customer_unique_id,
  SUM(revenue) AS total_spend,
  RANK() OVER (ORDER BY SUM(revenue) DESC) AS customer_rank
FROM master_data
GROUP BY customer_unique_id;
```

Result Grid  Filter Rows: Search  Export:

| customer_unique_id | total_spend | customer_ra... |
|---|---|---|
| 1617b1357756262bfa56ab541c47bc16 | 109312.640625 | 1 |
| bd5d39761aa56689a265d95d8d32b8be | 45256.0009765625 | 2 |
| be1b70680b9f9694d8c70f41fa3dc92b | 44047.998046875 | 3 |
| 05455dfa7cd02f13d132aa7a6a9729c6 | 36489.240234375 | 4 |
| 1ff773612ab8934db89fd5afa8afe506 | 30186.0009765625 | 5 |
| ec5b2ba62e574342386871631fafd3fc | 29099.51953125 | 6 |
| e7d6802668de6e74d0d6c56565bf2a24 | 22346.59912109375 | 7 |
| 8c20d9bfbc96c5d39025d77a3ba83d7f | 21874.050537109375 | 8 |
| f7622098214b4634b7fe7eee269b5426 | 19457.04052734375 | 9 |
| 71901689c5f3e5adc27b1dd16b33f0b8 | 19174.3798828125 | 10 |
| cb87122c4871e202777cf243fbea2d12 | 18666.99951171875 | 11 |
| 10de381f8a8d23fff822753305f71cae | 18384.750366210938 | 12 |
| 91f92cfee46b79581b05aa974dd57ce5 | 17786.8798828125 | 13 |
| 4a60b2ce1ee8c7b828e4bbcca5b86b41 | 17670.999755859375 | 14 |
| ed583a2a1eaf0dedc33af816153e8b8c | 17069.759765625 | 15 |
| 0c792d32a3251b4f69dae8646dfbedbc | 16313.599609375 | 16 |
| daf15f1b940cc6a72ba558f093dc00dd | 14963.6396484375 | 17 |
| 88324c93ce11436ae046563bf0da285c | 14577.56982421875 | 18 |
| 78fc46047c4a639e81ff65f0396e02fe | 14400.999755859375 | 19 |
| d5f2b3f597c7ccafbb5cac0bcc3d6024 | 14196.2802734375 | 20 |
| 6152fbfc8a92ee25fd821740bd33b089 | 12490.8798828125 | 21 |
| e2f063c52fa92b54f186c62b90699c10 | 12300 | 22 |
| 5c51fe11e0c3a016201653955c6090ee | 11832.18017578125 | 23 |

## CATEGORY REVENUE

```
SELECT
  product_category_name,
  SUM(revenue) AS total_revenue
FROM master_data
GROUP BY product_category_name
ORDER BY total_revenue DESC;
```

Result Grid  Filter Rows: Search  E

| product_category_name | total_revenue |
|---|---|
| cama_mesa_banho | 1692714.2811698914 |
| beleza_saude | 1620684.0394659042 |
| informatica_acessorios | 1549372.58648777 |
| moveis_decoracao | 1394466.9315271378 |
| relogios_presentes | 1387362.4486122131 |
| esporte_lazer | 1349446.9293460846 |
| utilidades_domesticas | 1069787.9702911377 |
| automotivo | 833745.6710395813 |
| ferramentas_jardim | 810614.9306926727 |
| cool_stuff | 744649.3201637268 |
| moveis_escritorio | 636124.8769683838 |
| brinquedos | 603564.3197097778 |
| bebes | 525553.2195739746 |
| perfumaria | 496263.5999403 |
| telefonia | 469890.57076358795 |
| papelaria | 308674.30956459045 |
| pet_shop | 307716.3603286743 |
| pcs | 274671.8794898987 |
| eletronicos | 245869.52997016907 |
| unknown | 240995.8801202774 |
| construcao_ferramentas_construcao | 237873.4802980423 |
| instrumentos_musicais | 221118.1608724594 |
| eletroportateis | 217024.01980400085 |

## STATE REVENUE

```
SELECT
  customer_state,
  SUM(revenue) AS total_revenue
FROM master_data
GROUP BY customer_state
ORDER BY total_revenue DESC;
```

Result Grid  Filter Rows: Searc

| customer_state | total_revenue |
|---|---|
| SP | 7403993.287906647 |
| RJ | 2688933.90042305 |
| MG | 2281229.1595630646 |
| RS | 1110976.4689769745 |
| PR | 1030822.3895778656 |
| BA | 773182.0207176208 |
| SC | 767093.9692153931 |
| GO | 493068.702003479 |
| DF | 421374.8593406677 |
| ES | 398321.89940071106 |
| PE | 361028.06939697266 |
| CE | 333406.17094039917 |
| PA | 254824.97995948792 |
| MT | 251504.48078155518 |
| MA | 195078.6299610138 |
| PB | 177370.58971977234 |
| MS | 162748.160490036 |
| PI | 133647.81045913696 |
| RN | 115084.5998954773 |
| AL | 108425.07027053833 |
| SE | 85694.32004356384 |
| TO | 70934.11979484558 |
| RO | 65200.66002655029 |

Used MySQL in this project and

- Ran KPI queries ✅
- Used window functions ✅
- Built analytical queries ✅